

Misogyny Detection in Internet Memes: Multi-Task Learning with Adversarial Training and RAG-Based Synthetic Data Generation

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Abstract

We present a text-only system for detecting misogyny in internet meme captions using the MAMI dataset from SemEval-2022 Task 5 (7,500 samples). Our approach combines a domain-adapted RoBERTa backbone (pre-trained on 124M tweets) with **multi-task learning (MTL)** to jointly solve binary misogyny detection and four-label sub-type classification in a single shared model. We apply **Fast Gradient Method (FGM)** adversarial training to improve robustness to OCR noise and lexical variation, and employ per-label threshold tuning instead of a uniform 0.5 cutoff. On 1,500 held-out test samples, our MTL+FGM system achieves macro F1 of **0.8253** (binary) and **0.5199** (multi-label, cascaded), surpassing a TF-IDF Logistic Regression baseline by +4.7% and +7.7% respectively. We complement the classifier with a complete **Retrieval-Augmented Generation (RAG)** pipeline using a locally quantized Mistral 7B model that generates type-distinguishable synthetic memes for evaluation. Error analysis reveals that 69% of failures are attributable to implicit or visually-dependent misogyny, with negation (11%) and irony (6%) as the leading linguistically identifiable failure modes.

1 Introduction

Online misogyny hostile or derogatory content targeting women spreads rapidly through internet memes, which combine visual humor with text to carry harmful stereotypes under the guise of a joke (Fersini et al., 2022). Automated detection is essential for content moderation at scale, but presents distinct NLP challenges: informal register, deliberate misspellings, irony, and meaning that often depends on the image rather than the text alone.

The **MAMI** shared task at SemEval-2022 (Task 5) formalizes this problem with two sub-tasks: (A) binary detection of misogynous content, and (B) identification of sub-types *shaming*, *stereotype*, *objectification*, and *violence*. This paper tackles both sub-tasks using a **text-only** approach on OCR-extracted captions.

The academic challenge is considerable: state-of-the-art text-only submissions to the SemEval-2022 shared task achieved macro F1 in the range

0.710.73 on Task A. Many harmful memes encode meaning through visual irony or cultural implicature invisible to any text model. Tackling this task therefore requires careful architecture choices, domain-matched pre-training, and explicit handling of class imbalance and threshold calibration.

Our main contributions are:

1. A **multi-task model** (shared RoBERTa backbone + two classification heads) that trains both tasks jointly, giving the multi-label head access to all 7,500 samples through shared representations $2.8\times$ more gradient signal than a separate model trained on the misogynous-only subset.
2. **FGM adversarial training** applied to hate speech classification, improving robustness to the lexical noise inherent in meme text (OCR errors, deliberate misspellings, leetspeak).
3. A **cascaded prediction pipeline** enforcing

the logical constraint that sub-types cannot be active unless the meme is first classified as misogynous, eliminating an entire class of structurally impossible prediction errors.

4. A full **RAG pipeline** (ChromaDB + BM25 + Mistral 7B) that generates synthetic memes used for distributional shift evaluation and for augmenting the training corpus.
5. Comprehensive **error analysis** with automatic failure categorization across five linguistically motivated failure modes.

The system is fully reproducible (all seeds fixed, Docker-containerized) and reaches production-quality code standards (pylint 10.00/10, 61 unit tests, ONNX export for framework-independent inference).

2 Related Work

Misogyny detection. SemEval-2022 Task 5 attracted 83 teams; the best text-only approaches reached macro F1 $\approx 0.710.73$ on Task A (Fersini et al., 2022). Top-performing multimodal systems leveraged ResNet or ViT image features alongside text, reaching F1 up to 0.79. Our text-only system at 0.8253 surpasses the best text-only submissions, demonstrating that careful model selection and training strategy can substitute for visual features on this dataset.

Domain-adaptive pre-training. Gururangan et al. (2020) demonstrate that continued pre-training on in-domain text consistently outperforms larger general-domain models across diverse tasks. The gains are particularly large for tasks involving informal language, abbreviations, and community-specific vocabulary all hallmarks of internet meme text. We apply this insight by selecting `cardiffnlp/twitter-roberta-base-hate`, pre-trained on 124M tweets and fine-tuned on HatEval and OfComEval, providing two layers of domain alignment: genre (social media) and task (hate speech detection).

Multi-task learning. Ruder (2017) surveys hard-parameter-sharing MTL, showing that auxiliary tasks act as regularizers by forcing the shared encoder to learn representations that generalize across related objectives. Caruana (1997) first formalized this inductive bias, demonstrating that jointly trained networks outperform single-task networks when tasks share latent structure. In our setting, the binary detection task provides dense gradient signal from all 7,500 samples to the shared backbone, while the multi-label head specializes the output layers. This is a natural fit because detecting misogyny and classifying its sub-type require the same semantic understanding of hostile language directed at women.

Adversarial training for NLP. Miyato et al. (2017) proposed perturbing word embeddings with gradient-based noise, achieving consistent improvements on text classification without any architectural change. The key intuition is that real-world text contains systematic noise: OCR errors, intentional obfuscation, creative spellings that single-point training ignores. We adopt FGM for its favorable accuracyefficiency tradeoff over multi-step methods such as PGD (Madry et al., 2018) and FreeLB (Zhu et al., 2020). FGM adds only one extra forward/backward pass per training step versus PGD’s K extra passes, reducing training time overhead from $K \times$ to $2 \times$.

Retrieval-Augmented Generation. Lewis et al. (2020) introduced RAG, grounding LLM outputs in retrieved evidence to reduce hallucination and enable knowledge-intensive generation. Reciprocal Rank Fusion (RRF) of dense and sparse retrievers (Cormack et al., 2009) improves robustness over either retriever alone by combining complementary signals: dense retrieval captures semantic similarity while BM25 excels at exact keyword matching. The ReAct framework (Yao et al., 2023) enables iterative, multi-step retrieval through a reasoningaction loop, allowing the agent to decompose complex queries into sub-queries and progressively refine its answer.

3 Dataset and Preprocessing

MAMI dataset. The Multimedia Automatic Misogyny Identification (MAMI) dataset contains 7,500 English meme captions annotated for: (i) binary misogyny; (ii) four sub-types: *shaming*, *stereotype*, *objectification*, and *violence*. The dataset was released for SemEval-2022 Task 5 and represents one of the largest manually annotated misogyny corpora in the literature.

Label distribution. Class balance is approximately 47%/53% (misogynous/non-misogynous) for the binary label mild imbalance manageable with class-weight rebalancing. Among the approximately 3,530 misogynous memes, sub-type distribution is substantially more skewed:

Sub-label	% of misog.	pos_weight
stereotype	61%	0.65
objectification	46%	1.19
shaming	27%	2.85
violence	21%	3.78

The `pos_weight` values (ratio of negatives to positives) are used as BCE class weights to counteract imbalance during multi-label training. Violence and shaming require the largest corrections, explaining their consistently lower recall in the baseline model.

Label co-occurrence. Sub-types are not mutually exclusive: 38% of misogynous memes carry two or more labels simultaneously. The most frequent co-occurrence pattern is stereotype+objectification (19% of misogynous memes), followed by stereotype+shaming (11%). Violence rarely co-occurs with other labels (4%), consistent with it representing a distinct, more extreme form of content. This co-occurrence structure motivates a multi-label formulation over a single 16-class categorical model.

Preprocessing. We deliberately apply **minimal preprocessing**: text columns are concatenated into a single `Text` field. No stemming, lemmatization, or stop-word removal is performed the BPE tokenizer of RoBERTa handles sub-word variation natively, and these features are informative signal in

the tweet domain. Usernames are preserved (they may indicate quote context), URLs are stripped (no semantic content in captions), and explicit Unicode normalization is skipped (the BPE vocabulary handles common variants). Captions are truncated to 128 tokens (covering 99.7% of the dataset without loss).

Data split. A single stratified split (stratified by binary label, `random_state=42`) is shared by both tasks to prevent cross-task leakage and ensure comparable metrics:

Partition	%	Samples
Train	70%	5,250
Validation	10%	750
Test	20%	1,500

Stratification ensures that both the binary ratio (47/53) and approximate sub-type frequencies are preserved across partitions. The test set is never used for hyperparameter selection, threshold tuning, or early stopping. Train/validation overlap is explicitly checked via SHA-256 fingerprinting of the `Text` column, confirming zero data leakage.

4 Methodology

4.1 Non-DL Baseline

We implement a cascaded TF-IDF + Logistic Regression baseline following the same two-stage architecture as the transformer model to ensure fair comparison. This also verifies that performance gains come from the model architecture and training strategy, not from any difference in the evaluation pipeline.

- **Stage 1 (binary):** TF-IDF (50k features, bigrams, `sublinear_tf`) + LR ($C=1.0$, balanced weights).
- **Stage 2 (multi-label):** `MultiOutputClassifier`(LR) trained exclusively on misogynous-only training samples.
- **Cascade:** Stage 2 predictions zeroed when Stage 1 predicts non-misogynous, enforcing the pipeline’s logical constraint.

Bigrams are critical for this task: patterns like *"not bad"*, *"pretty good"*, and *"totally belongs"* are informative negation/irony signals invisible to unigram models. The `sublinear_tf` weighting prevents high-frequency derogatory terms from dominating.

4.2 Transformer Model Selection

We evaluate three pre-trained models, selecting `cardiffnlp/twitter-roberta-base-hate` (RoBERTa-base, 125M params) based on both domain alignment and empirical results:

Model	Bin F1	ML F1	Decision
DistilBERT-base	0.750	0.400	rejected
DeBERTa-v3-base	0.740		rejected
twitter-roberta	0.821	0.500	selected

DistilBERT (66M params) lacks the capacity for nuanced hate-speech representations. DeBERTa-v3-base, despite having 50% more parameters than RoBERTa, shows training instability under FGM perturbations likely because its disentangled attention mechanism is more sensitive to embedding-space perturbation than standard RoBERTa. The Twitter-RoBERTa model provides: (i) tweet-domain vocabulary coverage (slang, abbreviations); (ii) pre-existing hate-speech feature extraction from HatEval and OffComEval fine-tuning; (iii) stable convergence under FGM.

4.3 Multi-Task Learning Architecture

MTL Model Architecture	
Input $\rightarrow$ RoBERTa backbone (768-dim [CLS])	
$\rightarrow$ Binary head:	
Drop(0.1)DenseTanhDrop(0.1)Proj(2)	
$\rightarrow$ ML head: Drop(0.1)DenseTanhDrop(0.1)Proj(4)	

Both heads output simultaneously from a single forward pass. The **joint loss** is:

$$\mathcal{L} = \mathcal{L}_{\text{binary}}(\text{all}) + \mathcal{L}_{\text{ML}}(\text{misogynous only})$$

$\mathcal{L}_{\text{binary}}$ is cross-entropy with class weights [1.0, 0.977] (ratio of negative to positive training

samples, compensating for the mild binary imbalance). $\mathcal{L}_{\text{ML}}$ is element-wise BCE with per-label `pos_weight` (Table 1). Restricting $\mathcal{L}_{\text{ML}}$ to misogynous-only samples prevents the model from learning trivially correct zero predictions for non-misogynous memes, which would suppress the gradients needed for sub-type discrimination.

Using a shared backbone gives the multi-label head access to gradient signal from all 7,500 samples $2.8\times$ more than training a separate model on the misogynous-only subset (2,655 samples). The Tanh activations in the hidden layers prevent gradient saturation more effectively than ReLU in the low-data regime of sub-type classification.

4.4 FGM Adversarial Training

At each training step, after the standard forward + backward pass, we apply one FGM perturbation step to the word embeddings:

$$\delta = \varepsilon \cdot \frac{\nabla_e \mathcal{L}}{\|\nabla_e \mathcal{L}\|_2}$$

with $\varepsilon = 1.0$. A second forward pass with perturbed embeddings $e + \delta$ accumulates additional gradients onto the same parameter tensors. Original embeddings are then restored before the optimizer step. This produces a model trained on both clean and adversarially perturbed inputs without any change to inference architecture or speed.

The perturbation simulates lexical noise (OCR errors, intentional obfuscation, creative spellings in meme text). For example, the word *"women"* is frequently written as *"womyn"*, *"wamen"*, or suffixed with derogatory terms in creative ways. FGM moves each embedding a fixed distance ε in the direction of maximum loss increase, encouraging the model to learn representations robust to embedding-space movements of this magnitude.

Compared to PGD (Madry et al., 2018), which applies K gradient steps (typically $K = 3$), FGM uses a single normalized step. This reduces training time overhead from $3\times$ additional forward/backward passes to $1\times$, making it practical for a 125M parameter model without a dedicated compute budget.

4.5 Cascaded Pipeline

At inference, the binary head gates the multi-label head: if `binary_pred = 0`, all sub-type predictions are forced to `[0,0,0,0]`, enforcing the logical constraint that sub-types are only meaningful for misogynous content. This eliminates an entire class of structurally impossible predictions (e.g., predicting *violence* but not *misogynous*) that otherwise inflate the false positive rate in the multi-label evaluation.

The cascade also improves multi-label F1 by focusing the Stage 2 decision threshold optimization on a purer set of positive examples the misogynous subset rather than the full test set where trivial true negatives would dominate the threshold calibration.

4.6 Per-Label Threshold Tuning

After training, per-label thresholds are optimized independently over the misogynous-only validation subset via grid search ($n = 80$ equally-spaced steps in $[0.1, 0.9]$, maximizing per-label F1). Tuning on misogynous-only samples avoids bias from the abundance of easy true-negatives in the full validation set.

Label	Threshold	$\Delta F1$ vs 0.5
shaming	0.465	+2.1%
stereotype	0.100	+5.4%
objectification	0.292	+3.8%
violence	0.586	+2.7%

The wide range (0.10.586) demonstrates that a uniform 0.5 threshold is substantially suboptimal. The extremely low stereotype threshold (0.10) reflects that the model assigns high confidence to stereotype predictions pushing recall very high at minimal precision cost which is appropriate given that stereotype is the dominant sub-type and missing it is costlier than a false positive in the moderation use case.

4.7 Optimization Details

Hyperparameter	Value
Optimizer	AdamW
Backbone LR	$3e-5$
Head LR	$3e-4$ ($10\times$ backbone)
Weight decay	0.01
Batch size	16 (physical)
Grad accumulation	4 steps (effective 64)
LR schedule	Cosine + 10% warmup
Gradient clipping	<code>max_norm = 1.0</code>
Precision	fp16 (AMP)
Early stopping	patience = 8 epochs
Val metric	$0.4 \cdot F1_{\text{bin}} + 0.6 \cdot F1_{\text{ML}}$

Discriminative fine-tuning (differential learning rates) allows the transformer backbone to adapt slowly preserving pre-trained representations while the randomly initialized classification heads converge faster at $10\times$ higher LR. Mixed-precision fp16 reduces VRAM usage by 40% and accelerates training by $1.7\times$ on GPU. All experiments use `RANDOM_STATE=42` for full reproducibility across `numpy`, `torch`, `random`, and `sklearn`.

4.8 RAG Pipeline

A complete Retrieval-Augmented Generation system complements the classifier for the meme generation task:

- **Ingestion:** PDF/DOCX parsing via `pdfplumber/python-docx`; fixed-size chunking (256 words, 32-word overlap) stored in MinIO object storage.
- **Indexing:** Dense retrieval via ChromaDB with `all-MiniLM-L6-v2` embeddings (384-dim, 22M params) + sparse BM25 index built over the same chunks.
- **Retrieval:** Hybrid **Reciprocal Rank Fusion (RRF)**: $\text{score}(d) = \sum_m \frac{1}{k+r_m(d)}$, $k = 60$. Top-5 chunks retrieved per query.
- **Generation:** Mistral 7B Instruct (Q4_K_M GGUF, 4.4GB) served via `llama.cpp` HTTP server. Temperature = 0.85, top_p = 0.95, repeat_penalty = 1.2.

- **ReAct agent:** Iterative SEARCH/ANSWER loop (max 5 steps) for multi-step queries, with explicit reasoning trace logged per step.

The pipeline runs entirely locally via Docker Compose (5 services: MinIO, ChromaDB, llama.cpp, Flask API, Streamlit). No external API calls are made, ensuring data privacy and eliminating API costs.

5 Experiments and Results

5.1 Main Results

Observations. Performance correlates with label frequency: stereotype (61% of misogynous, F1 = 0.64) > objectification (46%, 0.56) > shaming (27%, 0.45) > violence (21%, 0.43). Recall dominates precision across all labels a deliberate consequence of per-label threshold optimization (e.g., stereotype threshold = 0.10). This precision-recall trade-off is appropriate for a content moderation use case where missing harmful content is more costly than false positives.

For binary classification, the model achieves near-balanced P/R for both classes (Table 2), indicating that class-weight rebalancing was effective. The non-misogynous class achieves marginally higher F1 (0.83 vs. 0.82), consistent with the slight majority class advantage.

5.2 Ablation Project Evolution

The v1→v2 gain (+0.07 binary, +0.10 ML) is dominated by domain-matched pre-training confirming the importance of in-domain pre-training (Gururangan et al., 2020) for this task. The v2→v3 gain is more nuanced: the shared backbone transfers knowledge from all 7,500 samples to the multi-label head, producing the largest ML improvement (+0.020 absolute), while the binary gain is marginal (+0.004) since binary already had access to all samples in v2. FGM contributes approximately +0.003 binary F1 in isolation (measured by disabling FGM

while keeping MTL), showing that adversarial training provides a consistent, if modest, benefit.

5.3 RAG Meme Generation Evaluation

We generate 50 synthetic meme captions (10 per category: shaming, stereotype, objectification, violence, neutral) using the RAG pipeline and classify them with our trained model. Selected generation examples:

Category	Generated caption (sample)
stereotype	<i>“Of course she can’t read a map women don’t have a sense of direction.”</i>
objectification	<i>“The only thing she’s good for is standing in the background looking pretty.”</i>
shaming	<i>“Why does she dress like that and then complain about attention?”</i>
violence	<i>“Maybe if she talked less she would stay out of trouble.”</i>
neutral	<i>“Mondays are rough but coffee makes everything better.”</i>

The higher binary F1 on generated memes (0.8824) reflects that LLM-generated captions are more *explicit* than real memes, which rely on visual context and cultural implicature. The lower ML F1 (0.4291) reflects two factors: (i) generated memes are single-label by construction, diverging from the multi-label co-occurrence distribution in MAMI; (ii) small sample size (50) makes macro F1 high-variance. The RRF hybrid retriever successfully retrieved relevant MAMI training examples as context for each generation, verified by manual inspection of the top-5 retrieved chunks per query.

5.4 Error Analysis Failure Taxonomy

We apply automatic regex-based tagging to all misclassified test samples, identifying five failure categories based on surface-level linguistic features.

Key observations. (1) **OTHER dominates (6769%):** Most errors stem from *implicit misogyny* harmful content that requires cultural pragmatics,

world knowledge, or the meme image to interpret. Example: “*Just here to admire the scenery*” (posted with a photo of a woman) has no lexical signal available to the text model yet is annotated as objectification. These errors represent the fundamental ceiling of text-only approaches.

(2) SHORT_TEXT (1415%): Texts with <8 words provide insufficient context for the [CLS] representation to form a reliable decision. Examples: “*So true LOL*”, “*Exactly.*”, “*Goals*” each is inherently ambiguous without the image. Padding or retrieval augmentation could partially address this.

(3) NEGATION (1113%): The model sometimes focuses on content words while missing negation scope. Example: “*Women are definitely NOT bad at driving*” triggers a misogynous prediction due to co-occurrence of *women*, *bad*, and *driving* in the embedding neighbourhood. Negation-aware attention mechanisms or negation-specific data augmentation could help.

(4) IRONY/SARCASM disproportionately high in ML (9% vs. 6%): Irony is especially damaging for sub-type classification because it can simultaneously activate multiple wrong labels. Example: “*Oh sure, women totally belong in the kitchen said no functional society ever*” is sarcastic non-misogynous content, but contains strong stereotype and shaming lexical signals that confuse the multi-label head.

(5) AMBIGUOUS_HEDGE (12%): Hedged phrasing such as “*some would say...*” or “*people think that...*” creates uncertainty that the model cannot confidently resolve. These represent a small but distinctive failure class that could benefit from pragmatic discourse modeling.

6 Infrastructure and Reproducibility

6.1 Docker Architecture

The full system is containerized in a single `docker-compose.yml`. All services communicate via an internal Docker bridge network; only the Streamlit port (8501) is exposed externally.

Service	Image / Role	Port
minio	Object storage (chunks, models)	9000
chromadb	Vector store (dense embeddings)	8000
llama	llama.cpp (Mistral 7B GGUF)	8080
flask_app	REST API (classify + RAG)	5000
streamlit	Frontend UI	8501

6.2 API Surface

The Flask API exposes four endpoints:

- `POST /classify` accepts `{text: str}`, returns binary label and sub-type probabilities.
- `POST /generate` accepts `{category: str, n: int}`, returns `n` RAG-generated memes.
- `POST /search` accepts `{query: str, top_k: int}`, returns RRF-ranked chunks.
- `GET /health` service liveness check for Docker health probes.

6.3 Code Quality and Testing

Metric	Value
Pylint score (10.00/10.00)	10.00
Test suite (pytest)	61 / 61
Coverage scope	model, API, retrieval, ingestion
Code formatter	black
ONNX export	Yes (framework-independent)

All random seeds are fixed globally before any data loading or model initialization: `RANDOM_STATE=42` applied to `numpy.random.seed`, `torch.manual_seed`, `torch.cuda.manual_seed_all`, `random.seed`, and `sklearn.estimator.random_state` arguments. This ensures byte-for-byte identical results across machines with the same hardware-level floating point behavior.

The ONNX-exported model enables deployment without PyTorch: the `onnxruntime` inference engine runs on CPU with 12.3 ms/sample latency, making the system deployable on standard web servers without GPU hardware.

7 Qualitative Analysis

7.1 Confidence Score Distributions

Examining the sigmoid outputs before thresholding reveals that the model is well-calibrated on the binary task: the distribution of $P(\text{misogynous})$ is clearly bimodal with peaks near 0.1 and 0.9, indicating high confidence on most examples. The intermediate range $[0.35, 0.65]$ contains the majority of errors, as expected. For sub-type labels, distributions are more spread particularly for shaming and violence reflecting genuine ambiguity in the low-frequency labels.

7.2 Attention Pattern Analysis

Manual inspection of the top-attended tokens (from the final RoBERTa layer) on a sample of 50 correct and 50 incorrect predictions reveals two notable patterns:

- **Correct predictions:** Attention concentrates on domain-specific lexical cues (*kitchen, belongs, the real reason*) and sentence-final punctuation (exclamation marks, ellipses functioning as rhetorical signals).
- **False negatives (missed misogyny):** Attention is diffuse, often focusing on neutral frame words (*this, just, the*) when the harmful signal is in a multi-token idiom (*back to the kitchen*).

7.3 Synthesized vs. Real Distribution Shift

Comparing the prediction confidence distribution on real test set vs. RAG-generated memes confirms the distribution shift hypothesis: generated memes have a much higher proportion of high-confidence predictions (>0.90) than real memes (68% vs. 41%), confirming that LLMs produce lexically explicit content that is easy for text classifiers but does not reflect the full pragmatic complexity of real social media memes.

8 Limitations and Future Work

Multimodal extension. The largest failure category (69%) involves implicit misogyny that requires visual interpretation. The natural next step is a multimodal architecture combining the text encoder with a vision encoder (e.g., CLIP or ViT). Early experiments suggest that even a simple concatenation of RoBERTa [CLS] and CLIP image features improves recall on the OTHER failure category by an estimated 1015%, while the full system could approach the top multimodal SemEval-2022 submissions ($F1 \approx 0.79$).

Multilingual coverage. The MAMI dataset is English-only. Misogynous memes are a cross-cultural phenomenon and detection systems for other languages (Spanish, Italian, Arabic) would require both multilingual models (e.g., XLM-R) and language-specific annotations. Our pipeline architecture is language-agnostic; replacing the RoBERTa backbone with XLM-R would be the first step.

Temporal distribution shift. Internet slang and meme formats evolve rapidly. A model trained on 2022 data may degrade on 2025 memes due to new terminology, references, and visual tropes. Continual learning strategies or periodic re-annotation would be needed for a production deployment.

Computational cost. Training requires GPU access (902.6 seconds $\approx$ 15 minutes on an A100-class GPU). The quantized Mistral 7B model (Q4_K_M, 4.4 GB) runs on CPU but with 45-second generation latency, which is acceptable for batch generation but not for interactive real-time use.

Bias and fairness. The model inherits biases from the Twitter pre-training corpus, which overrepresents certain demographics, dialects, and cultural contexts. Predictions should be validated by domain experts before deployment, and algorithmic audits (e.g., performance disaggregated by demographic proxy variables) should be conducted.

9 Conclusion

We presented a production-ready NLP system for misogyny detection in internet memes combining (i) a domain-adapted RoBERTa backbone matched to the tweet register of meme text; (ii) multi-task learning for joint binary/multi-label training that multiplies the gradient signal available to the sub-type classification head; (iii) Fast Gradient Method adversarial training that simulates lexical noise at training time; (iv) a cascaded prediction pipeline enforcing logical consistency between the two tasks; and (v) per-label threshold tuning achieving +25% F1 per label over a uniform threshold.

The system outperforms the TF-IDF+LR baseline by +4.7% (binary F1: 0.787 $\rightarrow$ 0.825) and +7.7% (multi-label E2E F1: 0.483 $\rightarrow$ 0.520), surpassing the best text-only submissions to SemEval-2022 Task 5. Error analysis reveals that the dominant failure mode is implicit, visually-dependent misogyny a fundamental ceiling of text-only ap-

proaches affecting 69% of errors. Negation (11%) and irony (6%) represent the leading linguistically tractable failure modes and are natural targets for future work.

The complementary RAG pipeline (ChromaDB + BM25 + RRF + Mistral 7B) demonstrates that the same codebase infrastructure supports both discriminative classification and generative content synthesis at production quality, with zero external API dependencies and full local operation via Docker Compose.

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